Confidential Technical Feasibility Brief

HSLA-like Steel Architecture for Coupled Corrosion-Fatigue and Crack-Arrest Screening

Safe External Version v0.1 | For laboratory feasibility discussion only

CONFIDENTIAL DISCUSSION DRAFT: This document intentionally omits the underlying reasoning framework, internal scoring method, optimization logic, formula structure, weighting scheme, and any alloy recipe. It is intended only to request technical feasibility feedback from a qualified metallurgy laboratory.

1. Purpose

This brief asks whether a laboratory can evaluate, at coupon-level feasibility, an HSLA-like steel architecture that targets coupled naval / marine structural failure modes. The intent is not to request a complete alloy recipe at this stage. The intent is to translate a failure-chain hypothesis into a safe, testable laboratory screening plan.

The central question is: can a weldable HSLA-like matrix support both corrosion-fatigue-control behavior and crack-arrest behavior without degrading weldability, heat-affected-zone stability, or plate-scale manufacturability potential?

2. Scope and Non-Disclosure Boundary

This document is deliberately limited to the outer technical requirements. It does not disclose the internal framework used to derive the requirements.

Can be discussed	Cannot be disclosed in this version
Target failure chain and laboratory test gates	Internal reasoning framework, scoring logic, or formulas
Functional sample grouping concept	Exact optimization procedure or weighting method
Desired observable behavior: crack-arrest, corrosion-fatigue decoupling, HAZ stability	Alloy composition ranges, heat-treatment recipe, or proprietary design rules
Feasibility questions for laboratory review	Patent claim draft, military performance claim, or production-ready steel specification

3. Technical Problem Statement

Marine structural steels face coupled rather than isolated degradation. A useful early-stage architecture should therefore be evaluated against failure transitions, not only against single-property maxima such as yield strength, hardness, or standalone corrosion rate.

The target failure chain is:

marine environment -> localized corrosion / pitting -> cyclic fatigue loading -> pit-to-fatigue crack transition -> crack initiation -> crack-path connectivity -> accelerated crack growth -> weld / HAZ amplification -> structural failure risk

The proposed laboratory evaluation should focus on the following coupled transitions:

- Pitting-to-fatigue conversion: whether localized corrosion sites more easily become fatigue crack initiation points.
- Crack-path connectivity: whether microstructural / interfacial features encourage or interrupt connected crack propagation paths.
- Weld / HAZ amplification: whether welding thermal cycles create a preferential degradation zone for corrosion-fatigue or crack growth.

4. Proposed Functional Architecture

The proposed architecture is not a finished alloy. It is a screening direction:

HSLA-like weldable matrix + crack-arrest functional field + corrosion-fatigue-control functional field

The HSLA-like matrix is treated as the engineering anchor. Crack-arrest and corrosion-fatigue-control functions are treated as candidate enhancements that must not destroy weldability, HAZ behavior, or scale-up plausibility.

5. Suggested Coupon-Level Screening Groups

The laboratory may adjust the practical implementation based on available equipment, metallurgy route, safety constraints, and prior art. The following groups are conceptual and should not be interpreted as composition instructions.

Group	Conceptual role	Primary observation	Hard boundary
B0: Baseline	HSLA-like reference condition	Reference strength / toughness / corrosion-fatigue / weld response	Must be commercially or academically defensible as a reference
B1: CA-oriented	Crack-arrest / crack-deflection / crack-blunting emphasis	Evidence of interrupted crack connectivity or slower crack growth	Must not degrade weldability below reference viability
B2: CF-oriented	Corrosion-fatigue decoupling emphasis	Delayed pit-to-fatigue transition or reduced corrosion-fatigue sensitivity	Must not create brittle or HAZ-sensitive behavior
B3: CA-CF combined	Combined crack-arrest and corrosion-fatigue-control behavior	Best coupled-risk suppression without single-property collapse	Must preserve matrix-level manufacturability plausibility
B4: Optional SD later	Shock / dynamic energy dissipation exploration	Only considered if CA-CF stability is first shown	Must not override weldability, HAZ, or CF gates

6. Recommended Early Test Gates

The first laboratory goal should be gate-based feasibility, not final performance optimization.

Gate	Question	Preferred evidence type
G1 Weldability / HAZ stability	Does the candidate avoid obvious welding or thermal-cycle degradation relative to baseline?	Simulated HAZ, weld coupon, hardness / microstructure gradient, basic mechanical screening
G2 Corrosion-fatigue behavior	Does the candidate reduce pit-to-fatigue conversion risk or corrosion-fatigue sensitivity?	Pitting exposure + cyclic loading comparison, crack initiation statistics, surface / fracture analysis
G3 Crack-arrest behavior	Does the candidate interrupt crack connectivity, deflect cracks, blunt crack tips, or reduce growth rate?	Fractography, crack-growth tracking, microstructure-path correlation
G4 Coupled-risk balance	Does improvement in one function avoid degrading another critical function?	Multi-metric comparison against baseline rather than single best metric
G5 Scale plausibility	Is there any obvious barrier to plate-scale fabrication or repair-oriented workflow?	Laboratory expert judgment, processing complexity review, cost / process risk notes

7. Questions for the Laboratory

1. Is the HSLA-like matrix assumption reasonable for this type of coupled corrosion-fatigue / crack-arrest screening?
2. Which non-sensitive microstructural or processing mechanisms could plausibly support crack-arrest behavior while preserving weldability?
3. Which non-sensitive mechanisms could plausibly reduce pit-to-fatigue conversion or weld / HAZ corrosion-fatigue amplification?
4. Can the laboratory propose a coupon-level test matrix without requiring disclosure of a complete proprietary design method?
5. What minimum sample count, thermal-cycle simulation, corrosion exposure, fatigue method, and microscopy package would be needed for a credible first screen?
6. What obvious failure risks would make this direction unsuitable before any small-sample fabrication?

8. Success Criteria for a First Feasibility Round

A first feasibility round should be considered positive only if the combined CA-CF direction shows at least preliminary improvement in coupled-risk indicators while maintaining baseline viability. A sample with the highest strength but degraded HAZ behavior, corrosion-fatigue behavior, or crack connectivity should not be treated as the winning direction.

- Positive signal: improved corrosion-fatigue or crack-path behavior without obvious weld / HAZ degradation.
- Neutral signal: no improvement over baseline but no major degradation; the architecture may need redesign.

- Negative signal: single-property gain accompanied by weldability, HAZ, corrosion-fatigue, or manufacturability collapse.

9. IP and Collaboration Note

This brief is suitable for confidential technical discussion only. Before exchanging composition ideas, processing recipes, unpublished laboratory know-how, or sample results, the parties should consider an NDA or other appropriate written confidentiality arrangement. If the laboratory identifies a concrete implementable route and preliminary data are positive, patent counsel should evaluate whether protection is best pursued as a method, screening workflow, material architecture, processing route, or composition claim. This document itself is not a patent claim and should not be treated as legal advice.

10. One-Page Summary

This project is not asking for the strongest steel by a single metric. It asks whether a weldable HSLA-like steel architecture can be screened for lower coupled failure-chain risk: especially pit-to-fatigue conversion, crack-path connectivity, and weld / HAZ amplification. The most important first result is not a final alloy recipe, but a defensible feasibility answer: can crack-arrest and corrosion-fatigue-control functions coexist in a practical HSLA-like matrix without destroying welding and scale-up plausibility?